

ST. JOSEPH UNIVERSITY IN TANZANIA

ORDINARY DIPLOMA IN ENGINEERING PROGRAMMES (NTA LEVEL 6) - JULY 2026

SEM: II CET06202 MOBILE COMPUTING AND INTERNET OF THINGS

Time: 2:30 Hours

Max. Marks: 100

Centre Code: SJCET - 11

CODE: 119E22

PART - A (10 X 2 = 20 Marks)

Answer all questions

1. What is a sensor in an IoT system?
2. Define the term Machine-to-Machine (M2M) communication.
3. What is ZigBee technology?
4. State any two examples of wearable IoT devices.
5. Differentiate between cloud computing and fog computing.
6. State any two advantages of edge computing.
7. State any two challenges of interoperability in IoT systems.
8. Mention any two benefits of smart meters in IoT.
9. Mention any two common security threats in mobile computing.
10. Define the term Mobile Commerce (M-Commerce).

PART - B (5 X 4 = 20 Marks)

(Answer any five)

11. Describe the Internet of Things (IoT) and explain its architecture layers and how they interact to support smart systems.
12. Discuss the role of wireless communication technologies in mobile computing and IoT, including Wi-Fi, Bluetooth, and ZigBee.
13. Describe the role of mobile devices in communication systems and explain how cellular networks and Wi-Fi support mobility.
14. Discuss the challenges of security and privacy in IoT systems and explain possible solutions to overcome them.
15. Explain how IoT is used in smart cities and smart homes, highlighting four applications.
16. Explain how mobile devices integrate with IoT systems and the role of mobile applications in controlling IoT devices.
17. Discuss the future trends of IoT and explain how technologies such as AI, 5G, and Big Data will impact

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IoT development.

18. Explain how IoT is applied in healthcare, agriculture, and transportation systems, giving suitable examples.

19. Discuss the role of cloud computing in IoT systems and explain its advantages in storage, processing, and scalability.

20. Describe how IoT is used in industries and explain its impact on automation, productivity, and maintenance.